

Reference excerpt 01-011

Source: arXiv:1604.05171v1 | License: CC BY 4.0 | Word window: 356-615

gradient so that it is no longer constant over time and generates a non-steady ow [Sisini et al.(2016), Kalmanson et al. (1972)], ii) cyclically varying the CSA of the IJV [Sisini et al.(2015)]. Since the walls of the IJVs are neither rigid nor col- lapsed, the pressure variations generated in the RA are transmitted to the IJV as a pressure wave with nite 1

Figure 1: Geometrical model of the jugular-heart segment. The z axis is positive in the jugular-heart direction. l is the distance between the RA and point G in which the scan takes place. Blood velocity is indicated by w . propagation velocity c . Such pressure waves are responsible for the modulated component of the velocity of blood in the IJV. For this reason, their wave equa- tion can be used to derive the instantaneous velocity of the blood even in the absence of a direct measure- ment of such parameter obtained, for example, by using an ultrasound Doppler scanner. A rst attempt to this goal was presented in [Sisini et al.(2015b)], where the in- stantaneous velocity of the blood in the IJV was deter- mined qualitatively, for one cardiac cycle, using the Wom- ersley equation [Womersley(1955a), Womersley(1955b), Womersley(1955c)].

Physical description of the RA-IJV segment The RA-IJV system is represented in Fig. 1 as a tube with a circular section. G indicates a reference point on the right IJV. The $w(z, r, t)$ function represents the ve- locity of the blood in the z direction. It depends on z coordinate, on r (the distance from the